

RISHI MEHTA

ROBOTIC MANIPULATION | ROBOTICS HARDWARE | MULTIMODAL PERCEPTION | ROS2

College Park, MD | +1 240-349-4079 | rishi17@umd.edu | [linkedin.com/in/rishimehta17](https://www.linkedin.com/in/rishimehta17)

PROFESSIONAL SUMMARY

Robotics M.Eng. student with hands-on research experience in robotic manipulation, custom gripper development, embedded systems, and real-robot experimentation. Currently integrating a parallel-jaw gripper, Intel RealSense perception, and ROS2 control with a UR5 for surgical-tool pickup and sorting at Children's National Hospital. Experienced in PCB development, sensor and actuator integration, closed-loop control, Gazebo and MoveIt simulation, and hardware-software debugging. Research interests include VLA/VLM-based robotics, robot learning, multimodal sensing, and sim-to-real manipulation.

EDUCATION

M.Eng. in Robotics, University of Maryland, College Park (GPA: 4.00 / 4.00)

Expected May 2027

B.Eng. in Electronics and Communication Engineering, Gujarat Technological University, India

Jan 2025

TECHNICAL SKILLS

- **Embedded & Hardware:** STM32, ESP32, Arduino, Raspberry Pi, Jetson Nano, PCB design & fabrication
- **Fabrication & CAD:** SolidWorks, Onshape, 3D printing, rapid prototyping, mechanical assembly
- **Programming:** Python (NumPy, Pandas, Matplotlib), C/C++, MATLAB
- **Calibration & Testing:** Sensor calibration, encoder & BLDC motor calibration, hardware-in-the-loop testing, system debugging
- **Robotics:** ROS2, Gazebo, MoveIt!, OpenCV, URDF, ros2_control
- **Perception & Multimodal Sensing:** Intel RealSense RGB-D cameras, LiDAR, cameras, IMUs, optical encoders, ultrasonic sensors, current sensors, sensor calibration and data collection

EXPERIENCE

Robotics Research Intern

May 2026 - Present

Children's National Hospital | Washington, D.C. | Supervisor: Dr. Reza Monfaredi

- Developing and integrating a custom parallel-jaw robotic gripper with a UR5 manipulator for surgical-tool pickup and sorting experiments, incorporating Intel RealSense depth sensing for workspace perception and experiment monitoring.
- Building Python-based perception and control interfaces to enable closed-loop robot task evaluation and data collection.
- Developing a Vision-Language-Action pipeline for UR5 manipulator task execution, connecting perception inputs with robot-assisted sorting and pick workflows.

Research Assistant - IIT Gandhinagar Robotics Lab

Feb 2024 - Aug 2024

- Designed and fabricated a custom smart gripper for a 6-DoF manipulator, integrating current sensing, custom PCB electronics, and embedded actuator control.
- Developed a closed-loop control system in C++ and Python to regulate grip width and torque via real-time current feedback, enabling adaptive, repeatable grasping.
- Debugged hardware-software integration across mechanical, electrical, and firmware layers in a hands-on research lab environment for real-world prototype testing.

Software & Hardware Team Lead - GTU Robotics Club

Sept 2021 - Jan 2024

- Represented India at the international robotics competition ABU Robocon 2023 and led a 15-member software and hardware team (within a 40-person club) in design, fabrication, and programming of robots.
- Designed custom control loops and PCBs for sensor interfacing, motor driver circuits, and embedded microcontroller integration using STM32, ESP32, and Arduino.
- Calibrated SICK laser sensors, optical encoders, and BLDC motors; designed structural components in SolidWorks and Onshape and fabricated using 3D printing.

PROJECTS

Clutter Bot: Autonomous Mobile Robot | Raspberry Pi | Camera | IMU | Encoders | Ultrasonic Sensors

- Built and tested a fully autonomous DFRobot Baron-based mobile robot to identify, pick, transport, and drop RGB color blocks inside a 10 ft × 10 ft arena.
- Integrated Raspberry Pi, Pi Camera, IMU, wheel encoders, ultrasonic sensors, DC motor control, and a servo-actuated gripper into a complete hardware-software system.
- Implemented QR-based localization and mathematical pose correction to dynamically update robot position during autonomous arena operation.

Sketch-to-Motion: 6DoF Manipulator Trajectory Imitation | ROS2 | WebSockets | Gazebo | Python

- Developed a UR5 pipeline that converted user-drawn trajectories into inverse-kinematics-based robot motion with real-time ROS2 communication and visualization.